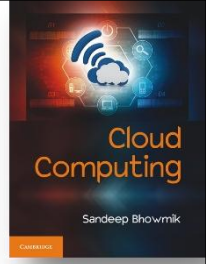
# Cloud Computing

Sandeep Bhowmik

## Chapter 3

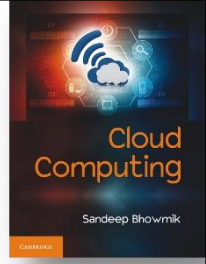
# Benefits and Challenges

Cambridge University Press



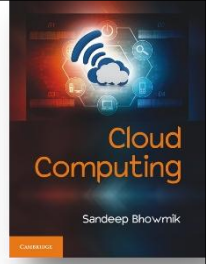

# The Transformation

- In the Cloud Computing era, users of computing are being addressed as Subscribers or Consumers.
- Computing is available in ready-to-use mode.
- One can subscribe to and consume any volume of computing (as well as computing resources) effortlessly.




# Initial steps towards Cloud Computing

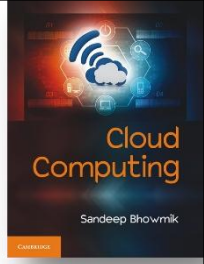
- 1999 - Initiative from Salesforce.com
  - The delivery of business (enterprise) applications via a 'normal' website was considered as the first-of-its-kind effort.
  - Salesforce's successful effort encouraged other software firms to deliver business applications via Internet.
  - This was the first step towards cloud computing.




# Initial steps towards Cloud Computing

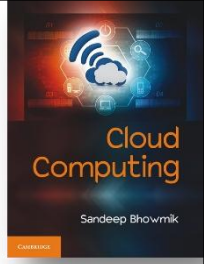
- 2002 - Initiative from Amazon
  - Amazon Web Service (AWS) was launched.
  - It delivered computing services over the Internet.
  - AWS provided a suite of services including storage.

- 2006 - Initiative from Google
  - ‘Google Docs’ introduced by Google brought cloud computing service to the public attention.



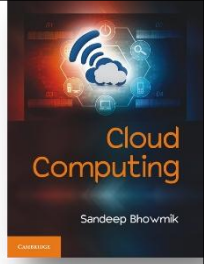

# Initial steps towards Cloud Computing

- 2006
  - Elastic Compute Cloud (EC2) introduced by Amazon.
  - Considered as the first concrete cloud service launched in the market.




# Utility Computing

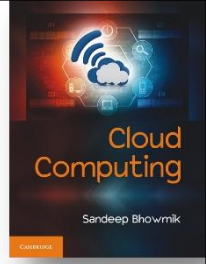
- It is the implementation of utility model of service delivery in the field of computing.




# What is Utility Service Model?

- A party arranges, owns, maintains and delivers some facility or service on-demand.
- This party is called vendor or provider.

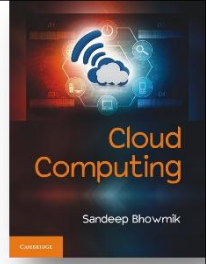
- Other parties can access the facility or service
  - from remote location
  - as and when required
  - on payment basis
- These parties are known as consumers or subscribers.




# What is Utility Service Model?

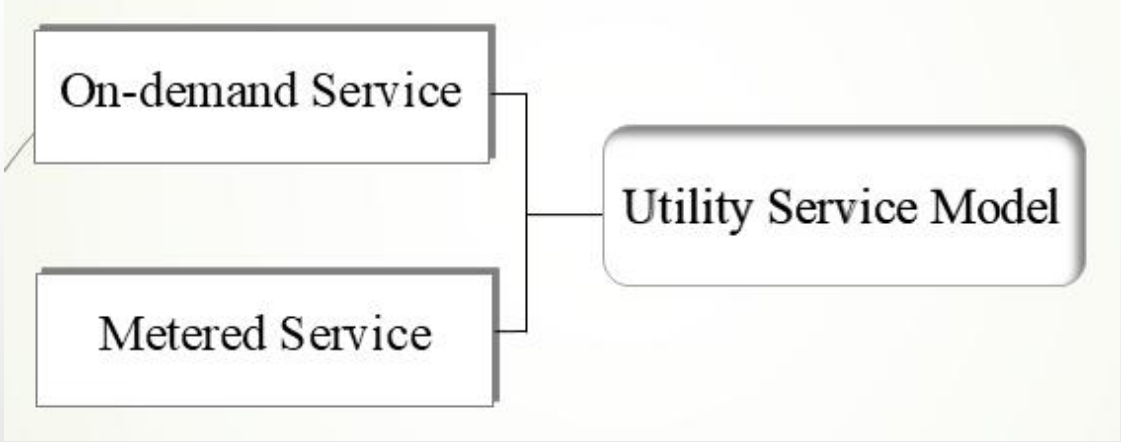
- The usages are metered at consumer's end.
- Bills are generated based on metered use.

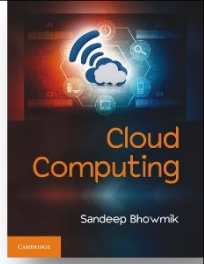
- The billing can be of two types:
  - Fixed rental
  - Actual use-basis




# What is Utility Service Model?

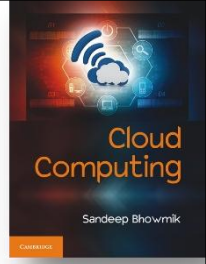
- Two aspects of utility model for service delivery






# Traditional way of computing vs. Cloud

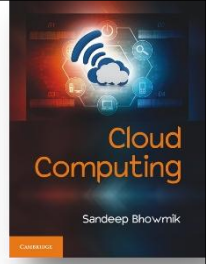
- Still doubtful about computing cloud adoption?
  - There is a similar century old story of electricity power generation and supply.
  - There, choice had to be made between maintaining own power generation unit, or subscribing to outside professional service.
  - And we know which philosophy has swept the market.




# Traditional way of computing vs. Cloud

- In computing also we have similar choices -
  - Either maintain own computing setup,

    or
  - Consume cloud computing service managed by more competent set of people.



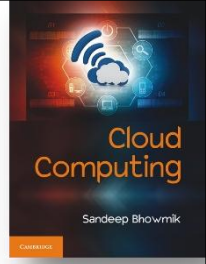

# Advantages in Cloud Model

## 1. The Utility Service Model

- Utility service model provides the facility of
  - very low initial investment
  - overall cost saving for subscribers

## 2. Metering and Billing

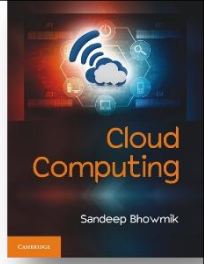
- Metering results in significant cost saving for users.




# Advantages in Cloud Model

## 3. Separation of Data Center Operation

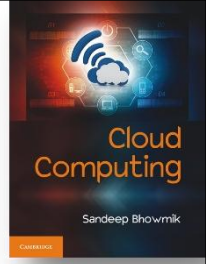
- Cloud vendor arranges and manages everything.
- It separates data center operations from activities at the users' end
- The users are relived of
  - building computing infrastructure
  - maintaining the system
  - running upgrades
  - etc.




# The Benefits

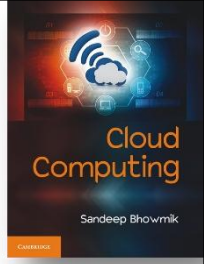
There are numerous benefits of cloud computing –

1. Less Acquisition/Purchase Cost
   - The initial investment of users adopting cloud computing is very low
2. Reduced Operational Cost
3. Reduced System Management Responsibility
4. Use-basis Payment Facility
5. Unlimited Computing Power and Storage
6. Quality of Service
7. Reliability

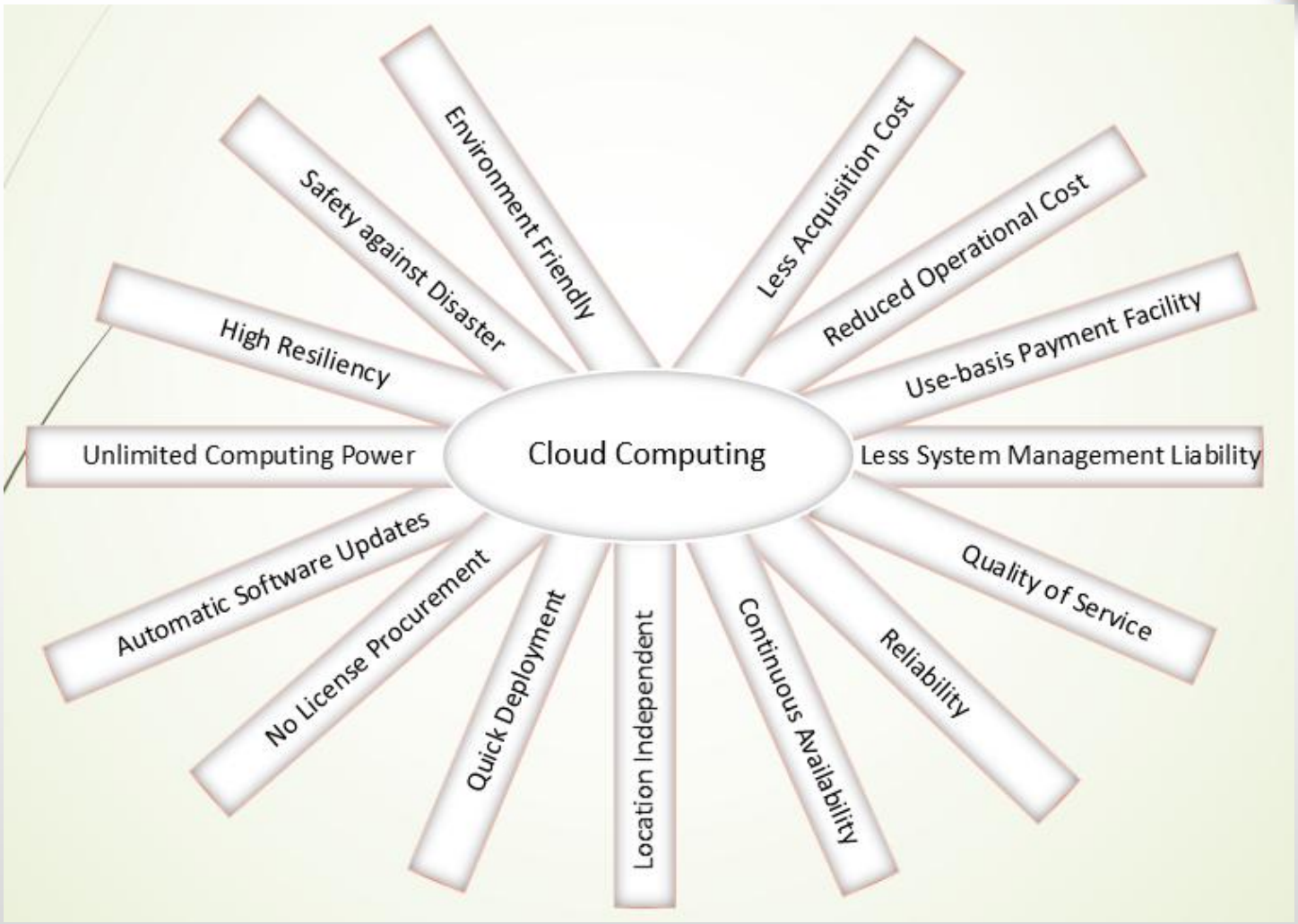


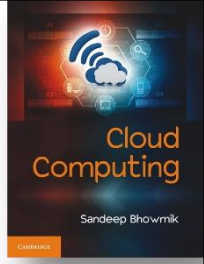

# The Benefits (Contd.)

8. Continuous Availability
9. Locational Independence/Convenience of Access
10. High Resiliency
11. Quick Deployment
12. Automatic Software Updates
13. No License Procurement
14. Safety against Disaster
15. Environment Friendly



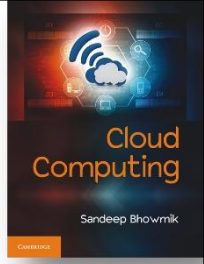

# The Benefits at a Glance





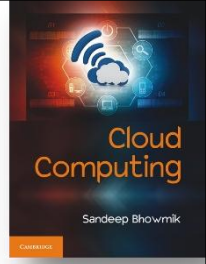

# The Challenges

- Limited Portability between Cloud Providers
- Inter-operability Problem
- Data Security
- Reduced Control over Governance
- Multi-Regional Compliance and Legal Issues
- Bandwidth Cost



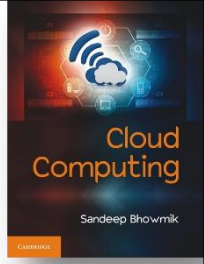

# Another Benefit

- Cloud computing promotes the idea of *ubiquitous computing*.
- Ubiquitous computing concept looks beyond personal computer like systems, and involves smart devices (anything with embedded electronic chips) to create network for better communication and data exchange.



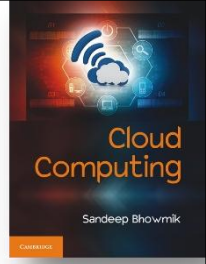

# How Cloud Computing Addresses Business Challenges

| Business Challenges | Cloud Computing Solutions |
|---|---|
| Budget | Provides flexibility by maximizing utilization of available budget, with the scope for future growth |
| Business | Reduced time to market provides scope for rapid business growth |
| Mobility of users | Enables anywhere access from a variety of devices |
| Flexibility | Agile nature of cloud computing enables easier and quick changes |
| Scope of Growth | Scalable (or elastic) architecture enables rapid growth |
| Availability | Provides almost 24×7 availability |
| Recovery | Integrates reliable backup and disaster recovery mechanisms |



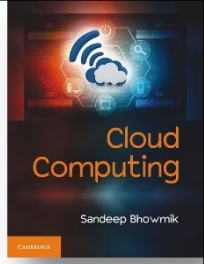

# The Ethical Issue

- Cloud computing philosophy is based upon a relationship of trustfulness between cloud computing vendor and its users.



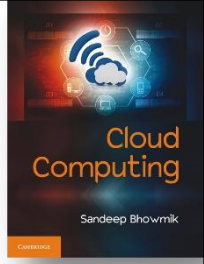

# Cloud - Network is the Computer

- Computing cloud comprises of a vast network of computing resources.
- The cloud is made of several heterogeneous computing systems spread over different geographic locations.
- The users visualize the entire thing as a single entity.



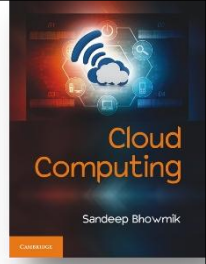

# The Challenge

- How applications running on those disparate systems communicate with each other?
- How processes/threads running on systems with different platform or architecture can communicate with each other?



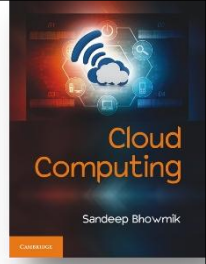

# The Solution

Web Service



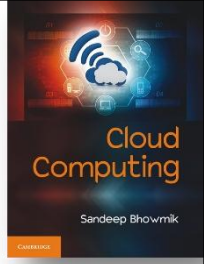

# Web-service

- Web service is the standardize way of establishing communication between two software systems over the internetwork.
- Due to this standardization, communication remains independent of
  - programming languages, or
  - platforms



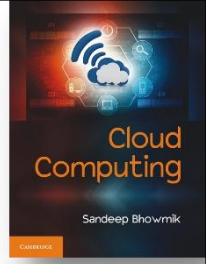

# Web-service

- Web service describes the method of establishing communication between two web-based applications.
- Web applications can communicate or publish their messages to the external world by using web services.



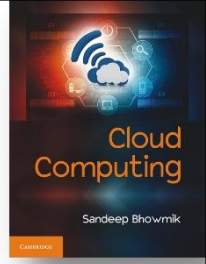

# Web-service

- Based on its implementation, a web service can either be
  - Simple Access Object Protocol (SOAP) based
  - Representational State Transfer (REST) compliant



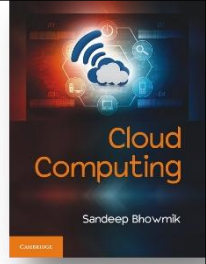

# SOAP & REST

- Both these approaches are useful and independent of
  - Computing architecture
  - Platform
  - Programming language



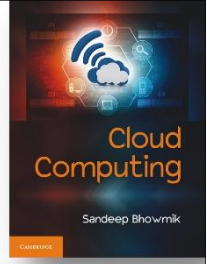

# SOAP vs. REST

| SOAP | REST |
|---|---|
| It is a message communication protocol that is used to build network application. | It is an architectural style for network application. |
| Uses XML for formatting message. | Uses simpler messaging formats like JSON. Also supports XML. |
| XML parsing required. | XML parsing can be avoided by using JSON. |
| Defines own additional security over HTTPS. | Inherits security measures from the HTTPS. |
| It is a bit heavy. | Lighter than SOAP, response time is better. |
| It is complex than REST. | It is simple to use and easy to understand. |



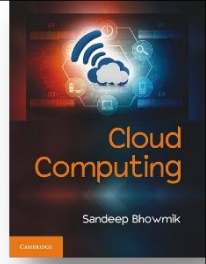

# SOAP vs. REST

| SOAP | REST |
|---|---|
| SOAP uses services interfaces to expose the business logic. | REST uses Uniform Resource Locator (URI) to expose business logic. |
| SOAP oriented development is faster since it is supported by many tools. | Development times is longer due to limited tool support. |
| Less flexibility in controlling resources. | More flexibility in controlling resources. |
| Requires less knowledge in programming. | Requires more knowledge in programming. |



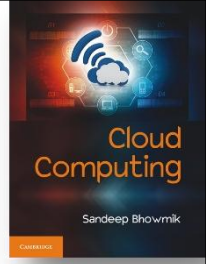

# SOAP & REST

- Both these approaches have been used by different cloud players.
- But, due to its simplicity, REST has gradually gained preference over SOAP in market.




# Cloud vs. Internet

- Two are different.
- Internet is a means for accessing cloud.
- Cloud computing is generally delivered via Internet.




# Thank You